

UTA027: Artificial Intelligence

L	T	P	Cr
3	0	2	4

Course Objective: This course introduces students to the fundamental concepts, techniques, and applications of Artificial Intelligence (AI). Students will gain theoretical knowledge and practical skills in areas such as problem-solving using search techniques, machine learning and designing intelligent agents for solving particular engineering problems.

Introduction to Artificial Intelligence: Foundations, scope, types of AI, problems, and approaches of AI

Intelligent agents: Structure of agents, Types of agent programs: reflex, model-based, goal-driven, utility-driven, and learning agents

Problem spaces: State Space Representation, Representation of problems as state space, problem characteristics, sample applications

Uninformed Search Algorithms: Brute Force search, Depth-First Search, Breadth-First search, Depth-Limited Search, Uniform Cost Search, Bidirectional Search

Informed search algorithms: Heuristic Functions, Best-First search, Beam Search, Hill Climbing, A* algorithm, AO graph, stochastic search algorithms: Simulated Annealing and Genetic Algorithm

Game playing: Minimax algorithm, alpha-beta pruning, iterative deepening

Introduction to Machine Learning: Well-Posed learning problems, Basic concepts, Designing a learning system, Types of machine learning: Supervised learning, Unsupervised learning, Semi-supervised Learning and Reinforcement learning, Types of data: structured and unstructured data.

Supervised Learning: Introduction to supervised learning tasks, Tree induction algorithms: split algorithm based on Information Gain (ID3), split algorithm based on Gain Ratio (C4.5), split algorithm based on Gini Index (CART), Instance based algorithms: K Nearest Neighbours (K-NN), Probabilistic algorithms: Naïve Bayes algorithm, Evaluation metrics

Unsupervised Learning: Introduction to supervised learning tasks, Partitioning-based methods

Laboratory Work (if applicable):

Basics of Python programming language: Data Types, Data Structures, Flow Control, Functions, Basic Data Science packages: NumPy, Pandas, SciPy

Implementing Search algorithms in C/C++/Java/Python: Depth first, Breadth first, Hill climbing, best first, A* algorithm, Implementation of games: 8-puzzle, Tic-Tac-Toe, tower of Hanoi and water jug problem using heuristic search

Implementing Machine Learning algorithms: Tree-based methods, K-NN, Naïve-Bayes algorithms, K-Means (from scratch and using sklearn library)

Course Learning Objectives (CLO)

The students will be able to:

1. Analyze methods and theories in the field of Artificial Intelligence and categorize various problem domains.
2. Design intelligent agents for concrete computational problems.
3. Analyze and apply different problem-solving strategies and search algorithms.
4. Implement and evaluate machine learning algorithms for various real-world tasks

Text Books

1. Russel S., Norvig P., Artificial Intelligence: A Modern Approach, Prentice Hall (2014), 3rd ed.
2. Murphy, Kevin P. Machine learning: a probabilistic perspective. MIT press, (2012) 3rd ed.

Reference Books

1. Rich E., Knight K. and Nair B. S., Artificial Intelligence, Tata McGraw Hills (2009) 3rd ed.
2. Luger F. G., Artificial Intelligence: Structures and Strategies for Complex Problem Solving, Pearson Education Asia (2009) 6th ed.